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RESEARCH ARTICLE

An Approach Toward Classifying Plant-Leaf Diseases and Comparisons With the Conventional Classification

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ABSTRACT Plants play a crucial role in human history by providing essential resources such as food. However, they are susceptible to various diseases like leaf blight, grey spots, and rust, posing significant challenges for farmers and ranchers due to the potential damage and financial losses incurred. The current method of diagnosing plant diseases relies heavily on manual inspection by trained professionals, which becomes increasingly complex and resource-intensive, particularly on large farms. To address this issue, a novel approach is proposed, integrating clustering methods and neural networks to enhance the efficiency of identifying and studying plant disease samples. This method aims to expedite the diagnostic process and accurately quantify disease-induced damage. It introduces a formula for assessing damage, calculating the diseased leaf area as a fraction of the total leaf area. With an impressive accuracy rate of 96–97%, the system successfully identifies diseases such as Alternaria, Anthracnose, Bacterial Blight, and Cercospora Leaf Spot. This advancement signifies a notable improvement over previous methods, highlighting the evolution of the system. By harnessing the capabilities of clustering techniques and neural networks, this innovative approach seeks to revolutionize the detection and monitoring of plant diseases in agriculture, offering a more accessible and precise solution.

INDEX TERMS Plants disease, clustering, neural networks, optimization, machine learning.

I. INTRODUCTION

Half of the global population relies on farming. This sets up the primary source of income. The role of agribusiness in India's public compensation is even more significant, so it's been said that farming in India could be the backbone of the Indian economy. The responsibility of farming for the first twenty years to get a stable public yield is somewhere between forty-eighth and an hour. In the years 2001-2002, this responsibility declined to effectively around twenty 6th [1]. The general share of agriculture and allied sectors, along with horticulture, prepared animals, and official organization and piscary sub pieces the degree that speed of significant worth is 13.9 % all through 2013-14 at 2004-05 costs [2]. Cultivation admissions contain a fifth of the entire tolls of the state [3], [4] as in Fig. 1.

Plant unwellness is a shortcoming of the ordinary state of a plant that obstructs or changes its fundamental limits. A wide range of plants, wild and grew the same, square measure powerless to unwellness. Though every money is powerless to reserve illnesses, these are, for each situation, modestly a couple [5]. The occasion and transcendence of plant sicknesses disagree from one season to another, dependent upon the closeness of the irresistible specialist, environmental conditions, and the harvests and varieties created. Some plant groupings square measure especially subject to scenes of

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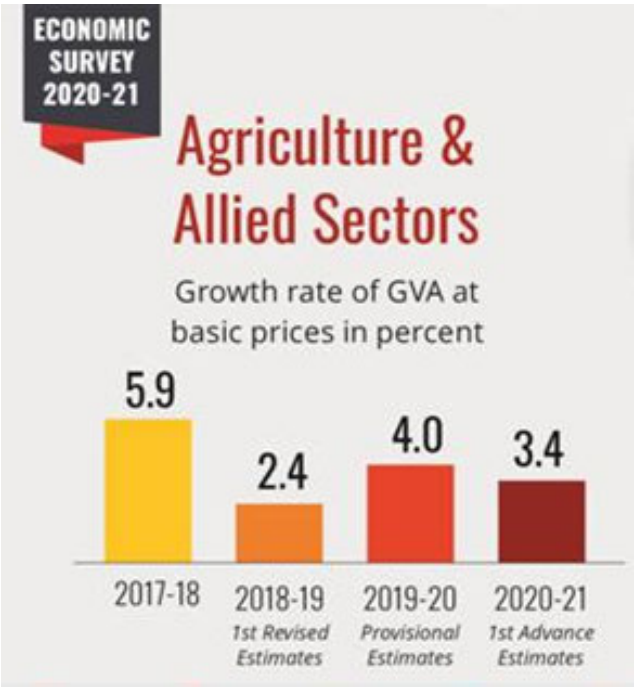


FIGURE 1. Agriculture and allied sectors [4].

illnesses, though others square measure increasingly more impenetrable to them

Plant illness is a problem with a plant's normal state that changes or stops it from doing what it is meant to do. Many plants, whether they grew wild or were grown in a garden, can't make you sick. Though every money is powerless to reserve illnesses, these are, for each situation, modestly a couple [5]. Plant diseases happen at different times and in other places each season. This depends on how close the expert is, how the environment is, and what crops and varieties are being grown. Some plant groups are especially susceptible to diseases, while others are becoming more and more resistant to them [6], [7]. See boot summation of plant infections.

Plant illnesses square measure is better known from times before the preeminent convenient structures. Fossil verification shows that plants were affected by unwellness 250 million years back. Loss of harvests from plant illnesses could moreover welcome on appetence and starvation, essentially in less-made countries any place admittance to infection prevention strategies is limited and yearly hardships of thirty to fifty percentage square measure customary for significant yields [8]. Genuine unwellness scenes among sustenance yields have provoked starvations and mass developments from the beginning of your time.

Plant illnesses square measure an ordinary piece of nature and one among fluctuated organic factors that work to keep the unnumbered living plants and animals in offset with each other. Plant cells contain extraordinary hanging pathways that overhaul their protection against bugs, animals, and microbes. One such model includes a phytohormone referred to as jasmonate (jasmonic corrosive) [8]. While not hurtful enhancements, jasmonate ties to uncommon proteins, referred to as JAZ proteins, to direct plant development, mud creation, and completely different procedures. Inside the sight of hurtful enhancements, be it because it could, jasmonate switches its drooping pathways, moving instead to coordinative procedures engaged with boosting plant barrier. Qualities that manufacture jasmonate and JAZ proteins speak to potential focuses for hereditary coming up with to deliver plant assortments with swollen protection from unwellness [9].

People have deliberately chosen and developed plants for nourishment, prescription, apparel, safe house, fibre, and magnificence for an outsized range of years. Unwellness is barely one among varied perils that have got to be viewed as once plants square measure far away from their regular home ground and developed in pure stands underneath what square measure of abnormal conditions.

Many important crops and ornamental plants are susceptible to disease and would have problems if people didn't take care of them. Developed plants are often more vulnerable to disease than their wild relatives. This is usually because a lot of the same species or types have the same genetic background and have grown up close to each other, sometimes over a considerable area of square kilometers. Under these conditions, it would be easy for an infectious agent to spread quickly. Most of the time, a plant becomes sick when it is constantly irritated by an agent that causes an abnormal physiological process that affects the plant's standard structure, growth, work, or other activities. This resistance with at least one of a plant's basic physiological or organic chemistry frameworks causes neurotic symptoms or side effects.

Plant diseases will be sorted in a big way based on the idea of their leading cause, which is either contagious or not, as shown in figure 2. Irresistible plant diseases square measure caused by an unhealthful creature, as an example, an organism, bacterium, mycoplasma, infection, viroid, nematode, or parasitic blooming plant [10]. Associate in Nursing irresistible operator is supplied for duplicating within or on its host and spreading beginning with one nerveless host and then onto following [11]. There are no diseases that can be passed from one plant to another that are caused by bad growing conditions. This includes temperature extremes, bad connections between water and gas, harmful chemicals in the soil or air, and too much or too little of an essential mineral. Since there don't seem to be any creatures that can correctly copy themselves within a group, there aren't any transmissible causative operators. This means that they can't be passed on.

Plant susceptibility to disease increases when they experience nutrient deficiencies or soil moisture imbalances. Prior infection by one pathogen enhances vulnerability to others. Disease complexity arises from the interaction of various factors. Understanding plant growth, variability, and environmental dynamics is crucial for disease identification [12].



FIGURE 2. Sample image of potato late blight.

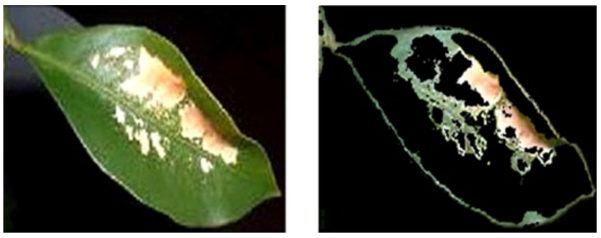


FIGURE 3. Image processing.

II. BACKGROUND

In this study, we investigate an innovative approach to classifying plant leaf diseases and contrast it with conventional classification methods based on the following steps.

1) Image Process: The standard approach to analyzing images involves using modern digital image processing techniques to eliminate noise and imperfections. Any disruptions or alterations in the image can occur either suddenly or gradually over time. Mathematically, an image can be represented as a two-dimensional function $f(x, y)$, where x and y are spatial coordinates, and the value of f at each point (x, y) corresponds to the intensity or brightness of the image at that location. When x, y, and intensity values are relatively small discrete numbers, the image is typically considered to be digital. In the past five years, significant advancements have transformed image measurement techniques, particularly in enhancing images captured by remote-controlled vehicles, aerial surveys, and military reconnaissance flights. These improvements are largely due to the increased accessibility of powerful computing devices, ample storage capacities, sophisticated design software, and other essential tools. Image measurement applications span various fields, including remote sensing, medical imaging, non-destructive testing, forensic analysis, textiles, materials science, military operations, industrial processes, document analysis, graphic design, and the printing industry.
2) Digital Image Process: ''Advanced image measurement'' refers to digitally processing a two-dimensional image. Tawde et al. [13] expand this idea to include any two-dimensional data. Complex images may only be fully understood through careful analysis. Digital image measurement methods offer flexibility, repeatability, and data accuracy preservation.
Image processing techniques are instrumental in plant disease diagnosis and management, enabling early detection and precise classification. Initial preprocessing refines raw plant leaf images by eliminating noise, correcting lighting, and enhancing contrast for clarity. Regular image updates ensure an accurate reflection of plant growth and disease progression, achieved through integrating new images with existing datasets. Subsequent segmentation of plant images into healthy and diseased areas facilitates focused analysis. Feature extraction algorithms discern key characteristics like lesion patterns or discolouration indicative of specific diseases. Consolidating information from multiple plant samples or regions through image gathering generates comprehensive datasets for training disease classification models. These techniques empower researchers and agricultural practitioners to detect, diagnose, and manage plant diseases effectively, bolstering crop health and agricultural productivity [1].
3) Image Preprocessing: In picture preprocessing, sensor data from satellites is refined to correct errors using mathematical models, as noted by Li et al. [14]. Picture enhancement involves adjusting brightness values to improve visual quality. In this process, specific image features are emphasized for analysis or display, including quality and edge enhancement, noise filtering, sharpening, and color enhancement. While enhancement doesn't add new information, it focuses on predefined image characteristics, as discussed by Gai et al. [15]. Enhancement algorithms vary based on application needs.
4) Image Segmentation: Image segmentation is crucial in image measurement, involving breaking down an image into distinct parts or ''objects.'' The level of segmentation depends on the problem's complexity, stopping when application goals are met. Thresholding divides a two-color image into object and background pixels, typically with objects as ''dark'' and background as ''white.'' Threshold selection methods vary, often based on establishing a balance across the entire image. In cases of uneven lighting or complex histograms, adaptive thresholding methods are used. Additionally, image segmentation can involve separating different regions within objects, with watershed segmentation being a common technique for this purpose.
5) Feature Extraction: Feature extraction methods are crucial in analyzing structured gap measurement system images, targeting high-level characteristics essential for target classification. These features encompass attributes like size, shape, composition, and spatial

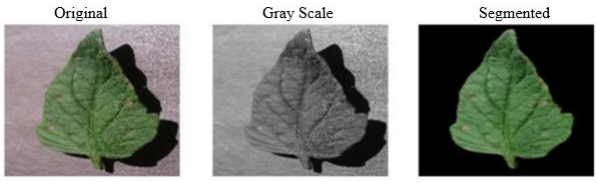


FIGURE 4. Image segmentation.

distribution, which define the objects of interest. Segmentation techniques isolate these objects, facilitating quantitative measurements for classification and description. Effective feature selection is paramount for achieving high confirmation performance, aiming to extract relevant information while minimizing variability. Choosing appropriate feature extraction methods tailored to specific requirements ensures optimal results across a wide range of applications.

6) Image Classification: The planned algorithm's effectiveness is evident in in-game results, particularly with the gearbox energy metric, demonstrating energy savings and prolonged system lifespan. As the algorithm's performance relies on limited future adjustments, it outperforms other energy-efficient algorithms. Image classification assigns labels to pixel groups based on intensity values, often requiring diverse features from multiple images [16]. Remote sensing utilizes multispectral characterization, either assisted or unassisted, for land use planning. Classification involves categorizing areas based on spectral characteristics, often using ghost credits from closed domains for training. Additionally, classification involves merging part data into spectral classes based on set criteria, utilizing features like structure, size, color, and texture, applicable even in magnetic resonance imaging [17].
7) Plant Severity Examination: Plant Severity Examination using Disease Detection on Leaves involves assessing disease severity by evaluating the damage inflicted on plant leaves. This process utilizes various tools and methods to identify diseases and determine their impact based on leaf symptoms and appearance. Initially, leaf samples are collected from potentially diseased plants and examined for visible symptoms such as spots, discoloration, lesions, wilting, or deformities, indicating the extent of disease progression. Advanced technologies like computer vision, machine learning, and image processing aid in disease detection and severity assessment by analyzing digital images of plant leaves. These methods employ algorithms to detect patterns, shapes, textures, and color changes associated with different diseases, estimating disease severity through comparison with databases or trained models. Accurate assessment of disease severity is crucial for implementing effective disease management strategies, including the use of fungicides, pesticides, or cultural practices like pruning or removal of infected leaves. It also enables tracking disease progression over time and evaluating the efficacy of management strategies. Overall, examining disease severity on plant leaves using detection techniques allows early detection, accurate diagnosis, and efficient management of plant diseases, contributing to improved crop health and yield.
8) Plant Severity Examination using ML: Machine learning, especially Multi-SVM (Support Vector Machine), is essential for finding plant diseases and figuring out how bad they are because it can do accurate and quick automated analysis.

   Early disease detection is a cornerstone of effective disease management strategies, particularly in the context of human health. Detecting diseases at their incipient stages allows for timely intervention, often resulting in better treatment outcomes and reduced morbidity and mortality rates. Various methods and technologies are employed in early disease detection, including diagnostic tests, screenings, and monitoring programs. These approaches aim to identify diseases before symptoms become apparent or before they progress to more advanced stages. Early detection not only improves individual patient outcomes but also contributes to public health initiatives by facilitating the prompt implementation of preventive measures and disease control efforts. Overall, early disease detection plays a crucial role in safeguarding public health and reducing the burden of illness on individuals, communities, and healthcare systems.

Combining the power of machine learning with Multi-SVM makes the process of finding plant diseases and figuring out how bad they are more accurate, efficient, and flexible. This technology could change the way crops are managed, cut down on losses, and help make agriculture more sustainable and productive.

A. MOTIVATION OF RESEARCH

The motivation driving this research stems from the urgent need to combat the profound challenges presented by plant diseases within agriculture. These diseases pose significant threats, resulting in substantial financial losses for farmers and jeopardizing global food security. Traditional methods of disease diagnosis are labor-intensive and time-consuming, often relying on manual visual inspection by trained experts. This research seeks to revolutionize this process by harnessing the power of image processing and machine learning techniques. By automating disease detection, the aim is to provide farmers with a rapid and accurate means of identifying and managing plant diseases, thus minimizing crop losses and ensuring a stable food supply. Additionally, early detection facilitated by this approach can enable timely intervention, reducing the spread of diseases and promoting sustainable agricultural practices. Ultimately, this research endeavours to contribute to the resilience and sustainability of farming systems worldwide

by providing innovative solutions to combat plant diseases effectively.

Contribution of our research: The contribution of this research lies in the development of a comprehensive approach for plant disease detection that integrates advanced image processing techniques and machine learning algorithms. By employing methods such as color segmentation, clustering, and conversion to Lab color space, the research offers a systematic framework for analyzing leaf images and identifying disease patterns. Furthermore, the utilization of a Multiclass Support Vector Machine (SVM) for disease classification enhances the accuracy and reliability of the detection process. This integrated approach enables the automatic and rapid identification of plant diseases, thereby reducing the reliance on manual inspection and providing farmers with a valuable tool for early disease detection and management. The research not only advances the field of agricultural science but also offers practical solutions to address the challenges posed by plant diseases, ultimately contributing to the sustainability and productivity of agricultural systems.

B. RESEARCH QUESTIONS

1) Question 1: How can advanced image processing techniques be optimized and integrated with machine learning algorithms to improve the accuracy and efficiency of plant disease detection and classification?''
2) Question 2: What is the comparative effectiveness of different clustering and segmentation methods in identifying diseased regions within plant leaf images, and how do these methods impact the overall performance of disease classification models?
3) Question 3: How does the integration of multiclass SVM with grid search optimization enhance the precision and reliability of plant disease classification, and what are the optimal hyperparameter configurations for maximizing classification accuracy?

These research questions aim to delve into various aspects of the proposed approach, including the optimization of image processing techniques, the selection of appropriate clustering and segmentation methods, and the refinement of machine learning algorithms for accurate disease.

III. LITERATURE REVIEW

It is essential to know how to spot plant diseases so that crops retain yield and quantity. The study of plant diseases looks at patterns in plants that can be seen. For agriculture to be sustainable, it is essential to keep an eye on plant health and find diseases. It is hard to keep track of plant diseases by hand. Image processing is used to find plant diseases. This takes a lot of work, knowledge of plant diseases, and a lot of time to process. Image acquisition, pre-processing, segmentation, feature extraction, and classification are the steps that make up disease detection.

A. RELATED WORK IN PLANT SEVERITY EXAMINATION USING IMAGE PROCESSING

It has been stated by Pawar et al. [18] that the absence of timely detection of diseases in crops or the failure to receive timely information about their presence indicates a dangerous situation. For this reason, and to gain knowledge of the disease in the crop, image processing technology can be employed by using steps such as image acquisition, preprocessing the image, segmenting the image samples, extracting features, detecting plant diseases, and classifying them.

In their paper Aithal et al. [19], the authors draw attention to how Agriculture is a cornerstone of the Indian economy, with 65% of people being reliant on its output. Disease detection plays a critical role in maintaining the quantity and quality of agricultural products. Symptoms of these diseases present themselves through leaf changes in color and the appearance of spots. However, manual methods for this process can be tedious and expensive. Thus, this research proposes an automated software-based solution to classify plant leaves for diseases using image processing. Upon upload, farmers can push 'predict' and have access to disease type, description, symptoms, and instructions for treatment or fertilization for the diseased leaf. This approach aims to detect early stages of illness so that appropriate measures can be taken; experiments have been done on grape, potato, and tomato crops from the Plant Village Database to demonstrate efficacy. By adopting such methods, productivity could be improved significantly.

Johnson et al. [20] highlighted the critical role of agricultural productivity in the nation's economy. According to a survey by India's linked chambers of commerce and industry, plant disease leads to an estimated fifty thousand crore rupees in annual agricultural-related losses and contributes to the starvation of two hundred million Indians. Thus, recognising illnesses among plants is essential for reducing food waste and boosting output with quality and quantity. Although traditional methods can be applied, they often require manual investigation of leaf patterns and disease spotting and necessitate large-scale surveillance by numerous personnel. With this in mind, several innovations have been introduced recently to improve production in the agrarian sector. Plants are an indispensable source of energy but can become vulnerable when diseases are present, leading to financial losses as well as social impacts. However, specific symptoms may be apparent upon initial inspection if undetected swiftly; a major setback could be imminent. This research study proposes a novel technique using Image Processing (IP), which is adept at discovering plant diseases by following a sequence involving image acquisition, pre-processing, segmentation, feature extraction, and classification. The histogram of an oriented gradient is used to extract features from an image of the affected area.

Aarizou and Merah [21] stated that in recent years, employing deep Convolutional Neural Networks (CNNs)

for the recognition of plant diseases from leaf images has become a heavily researched area that produces remarkable results. Still, training these classifiers requires datasets in sizeable amounts. The PlantVillage dataset, extensively used by scientists, is available without charge. However, the images are confined to laboratory pictures, which weakens the classifier's performance when working with complex field photos. In this paper, we look into using both field and laboratory images to train our deep CNN on identifying healthy and unhealthy plants. To construct a dataset consisting of 54000 images divided into 'healthy' and 'unhealthy' categories, we combine photos from the PlantVillage and EdenLibrary datasets. This collection is used to hone three state-of-the-art image classifiers pre-trained on the ImageNet dataset: AlexNet, ResNet34, and DenseNet121. The outcome indicates that utilizing the combined dataset enhances accuracy significantly for complex images.

Ramyashree et al. [22] argued that agricultural products are a fundamental necessity in every country, highlighting the devastating effects of plant diseases on farm output and economic assets. In line with prior research, 68% of agricultural productivity has been lost due to rising leaf disease incidences. To tackle this dire issue, this study implements a screening strategy to diagnose illness from taken photos, with plants such as potatoes, peppers, and tomatoes selected for analysis. These were subjected to preprocessing, image segmentation, and feature extraction; the K Nearest Neighbor (KNN) classification was then applied based on the results of these steps, yielding an impressive accuracy rate of 98.23% in predicting plant leaf diseases across all data sets.

Singh et al. [23] stated that Climate change is causing diseases in crops, resulting in lower agricultural production and impacting the agricultural financial system. To mitigate damage from these diseases, modern technologies such as artificial intelligence (AI), machine learning (ML), and deep learning can be used to identify their source. G.S. Singh et al. proposed an application that allows farmers to identify plant diseases by scanning leaves or uploading images of them. Image processing is accomplished using Convolutional Neural Network (CNN), Visual Geometry Group 16 (VGG-16), DenseNet-121, and Inception V3 algorithms which extract characteristics such as color from the photograph.

Sritharan, et al. [24] demonstrated an innovative solution to the problem of plant diseases affecting farmers globally with limited resources available online to address the issue. In Sri Lanka, they did a study that led to the creation of a mobile app that uses image processing to suggest pesticides for diseases. As part of the method for finding diseases, this included getting the images, pre-processing, segmenting, extracting features, and classifying them. In this research paper, CNN algorithms were also used to pull out features. The results showed that deep convolutional neural networks that had been trained and fine-tuned with a database of plants and leaves collected independently for specific plant diseases could be used to diagnose plant diseases. When it came to spotting plant diseases, the study was very accurate, with a 98 percent success rate. The mobile system was then used to suggest the suitable pesticides.

Alyas and Mohammed [25] this article aims to present different image processing and machine learning techniques that can be used to detect plant diseases based on images. We give information about the size of the photo collection, preprocessing, segmentation techniques, classifier types, classifier resolution, and other factors. Also, we explain how our study helps us set up our work on classifying and identifying plant diseases. Then, machine learning techniques are used to separate the main technical solutions used in the process. Support vector (SVM) and k-mean clustering techniques were used to get color and texture information from plant disease samples so they could be analyzed more thoroughly. The test results show that the SVM classifier does a great job of finding and identifying plant-borne diseases in crops.

Kundu et al. [26] We are all aware that without civilization, we cannot imagine a world with Agriculture. India is renowned as an agricultural nation, mainly due to its economy being heavily reliant on grains. This sector constitutes the foundation of any country's economy and provides a necessary means for citizens to maintain their livelihood and life quality. Nowadays, the increasing population has led to a higher demand for food, which is primarily fulfilled by Agriculture. In addition to satisfying this need, the sector also bolsters international relations between countries and helps develop their strength. A primary concern raised, though, is the purity of crops grown, are they accessible from harmful chemicals such as pesticides or insecticides? Thus, I have written this research paper to provide insights into techniques to detect diseases in plants through image processing so consumers can be better informed about what they eat and what risks are involved in consuming contaminated products.

Aiswarya and Sreekumar [27] provides an automated way of generating quantum images and converting color pixels into well-defined structures. The same idea of energy transformation is utilized in EDQRCI, where 2D mapping is used to encode values of energy into a set of quantum data. When electromagnetic data is available, it will be pre-processed and stored, then encrypted into a quantum mode to represent colour data. On the other hand, when infrared radiation is given, it can be encoded into colour representation based on the intensity of associated radiation. Lastly, Different Quantum Image Description makes it possible to represent, develop, and acquire data individually in quantum mechanisms for further experimentation purposes.

Dubey et al. [28] emphasize that image processing has become an integral part of agricultural research and progress. In particular, there are a lot of studies going on right now to find plant diseases. With this information, farmers can not only get the most out of their crops, but they can also do a wide range of farming tasks better. This paper

is mainly about making an algorithmic programme that uses machine learning techniques and image recognition tools to find and classify plant diseases. First, the affected area is found and recorded, and then the image is ''pre-processed.'' Then, fragments are collected to help with identification, and then features are taken from the infected area. This article also talks about leaf photography as a way to accurately find plant diseases. It also talks about different segmentation and extraction algorithms that can help find diseases.

Soni et al. [29] highlighted that crops act as the backbone of agricultural fields yet can fall prey to diseases, leading to reduced yield and quality. Finding and naming plant pathogens by hand is a complex and time-consuming job that requires expertise and costs more money. So, they came up with a plan for an automated system that could find crop diseases early on and reduce the amount of work in large fields. The model was made to quickly and accurately find leaf diseases on cucumber, guava, groundnut, pumpkin, and cowpea plants (98.27% accuracy based on 406 images of leaves). MATLAB was used during the development process for Pre-Processing, Thresholding, Sobel filtering, Morphological Dilation, and ROI Extraction.

In this paper, Yeh et al. [30] present an intelligent plant disease detection system that is based on the modified Alexnet convolutional neural network (CNN). Alexnet is very good at figuring out what's in a picture. However, figuring out what's wrong with a plant by looking at its leaves is more complicated than figuring out what's in a picture in general. So, Alexnet has been changed so that it can recognize plants of different sizes and angles and be used in the real world. The models are then trained using a training corpus pulled from the internet and graded by humans. Based on the results of their tests, the proposed method works very well.

Kamala and Anna Alex [31] pointed out that apples can be produced using the hydroponic system and that those grown this way tend to be sweeter than those cultivated in soil. Also, the first method is a lot more efficient when it comes to harvesting. Using Image Processing or Machine Learning, you can find diseases in apple fruit, which is essential for successful production, whether you're trying to make more, make better, or make more money.

Akter et al. [32], Agriculture has a significant impact on the GDP of a country and is an integral part of the economy. Plant diseases are a big problem in agriculture right now. They pose a significant threat to food security around the world and cause farmers to use too many chemicals and pesticides, which is terrible for the environment. If these diseases are found and named early on, it can be good for both the environment and the productivity of agriculture. This paper describes a way to tell the difference between healthy and unhealthy plant leaves and find diseases on unhealthy leaves that are easy to compute. Our method for classifying things is based on a Support Vector Machine (SVM), and we use Grid Search to find the best solution.

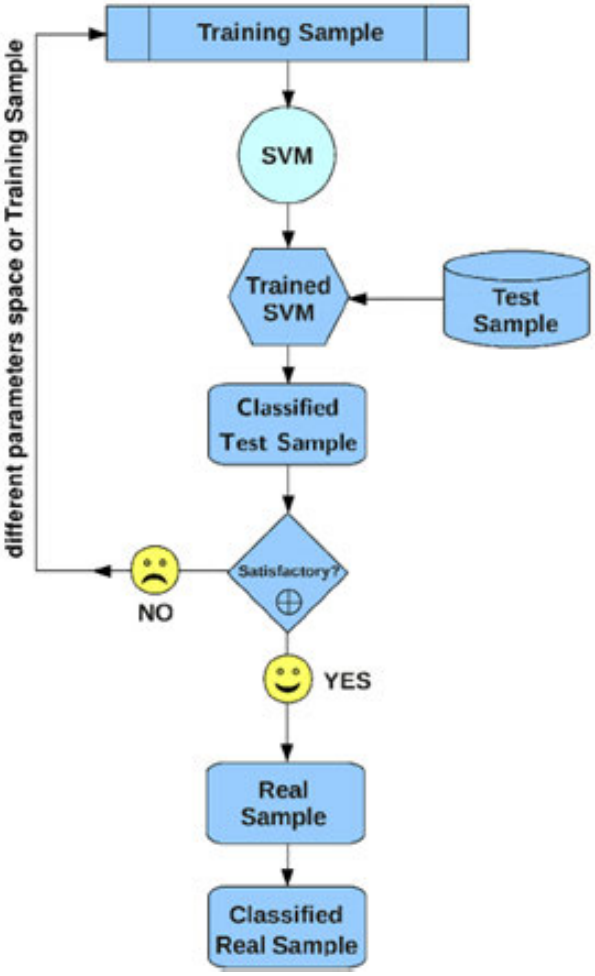


FIGURE 5. Image segmentation.

IV. OBJECTIVE OF RESEARCH

The existing method for detecting plant diseases is by expert naked-eye observation. When we do this on large farms, it requires a large team of experts and continuous monitoring of the plant. Use the following formula to calculate the percentage of disease damage: Diseased leaf area ÷ Entire leaf area × 100. The proposed approach will also use the clustering approach and the neural networks concept to examine the samples and detect diseases.

V. METHODOLOGY

A. SUPPORT VECTOR MACHINE (SVM

It's a regulated learning framework utilized for arrangement and relapse issues. Support vector machine is incredibly preferred by numerous individuals as it produces striking accuracy with less calculation power.

It is for the most part utilized in grouping issues. We have three sorts of learning directed, solo, and support learning. A support vector machine is a specific classifier officially characterized by partitioning the hyperplane.

1) **Advantages of SVM:** The support vector machine works similarly well when there is a reasonable edge for separation between the classes.

a) It is more useful in high-dimensional spaces.
b) It is compelling on occasions where the quantity of measurements is larger than the number of examples.
c) Support vector machine is equivalently memory methodical.

2) **Disadvantages of SVM:** Support vector machine algorithm isn't adequate for huge informational indexes.
   a) It doesn't execute very well when the informatio nal-based collection has more of the solid, for example, the target classes that are covered.
   b) In such situations where the quantity of the properties for every information point exceeds the quantity of the preparing information examples, the support vector machine will then fail to meet the expectations.
   c) Also, the support vector-based classifier works by the process of putting information focuses, one that is above and underneath the grouping-based hyperplane where there is no such probabilistic explanation for the arrangement.

B. K-MEANS CLUSTERING

K-means bunching is one of the simplest and most well-known algorithms for learning on your own. Most of the time, unaided algorithms conclude datasets using only information vectors and not referring to known or marked results, as shown in Fig. 6.

A ''cluster'' is a group of information focuses that have been put together because they are similar in some way. In information mining, the K-means algorithm starts with the first group of randomly chosen centres, which are used as the starting points for each group. It then uses iterative (monotonic) calculations to streamline the locations of the centres.

1) It ends up making and enhancing groups when, by the same token:
2) The centroids have balanced out — there is no adjustment of their qualities because the grouping has been fruitful.
3) The characterized number of cycles has been accomplished.

How the k-means algorithm works is as per the following:

1) Specify the number of groups K.
2) Initialize centroids by first rearranging the dataset and afterward arbitrarily choosing K information focuses for the centroids without substitution.
3) Keep emphasizing until there is no change to the centroids. i.e., the task of information focusing on groups is staying the same.
4) Compute the amount of the squared distance between information focuses and all centroids.
5) Assign every piece of information to highlight the nearest group (centroid).
6) Compute the centroids for the groups by taking the normal of all information focuses that have a place with each bunch.''

1) APPLICATIONS OF K-MEANS

The K-Means algorithm is very popular and is used for a wide range of things, such as market segmentation, report grouping, picture segmentation, picture pressure, and so on. Most of the time, when we do a group investigation, the goal is to:

1) Get a significant instinct for the design of the information we're managing.
2) Cluster-then-anticipate where various models will be worked for multiple subgroups if we accept there is a wide variety in the practices of different subgroups. An illustration of that is bunching patients into various subgroups and constructing a model for every subgroup to foresee the likelihood of the risk of having a coronary episode.''

C. MULTI CLASS SVM

In the field of machine learning, multiclass or multinomial grouping is the problem of putting events into one of three classes. (Twofold grouping is the process of putting social events into one of two groups.) Some sorting algorithms usually let you use grouped classifications, while others are generally just double-checked. These can still be turned into multinomial classifiers through a variety of methods. Multiclass grouping should be distinct from multi-mark characterization, where different names are expected for each event.

D. IMAGE SEGMENTATION

Image Processing, or, all the more explicitly, Digital Image Processing, is a cycle by which a digital image is handled utilizing a bunch of algorithms. It includes basic level tasks like clamor expulsion to everyday functions like distinguishing objects, individuals, text, and so forth, to more muddled tasks like image classifications as in Fig. 7, feeling detection, oddity detection, division, and so on A digital image is comprised of different parts that should be ''examined,'' how about we utilize that word for effortlessness sake and the ''examination'' performed on such parts can uncover a great deal of concealed data from them. This data can assist us with tending to plenty of business issues, which is one of the many ultimate objectives that are linked with image processing. Image Segmentation [33] is the interaction by which a digital image is parcelled into different subgroups (of pixels) called Image Objects, which can diminish the intricacy of the image and, accordingly, dissecting the image becomes less complicated.

With these names, we can set limits, set boundaries, and separate the most essential parts of an image from the rest of the less critical parts. In the model below, we tried to get the essential parts, like the seat, table, and so on, from a basic

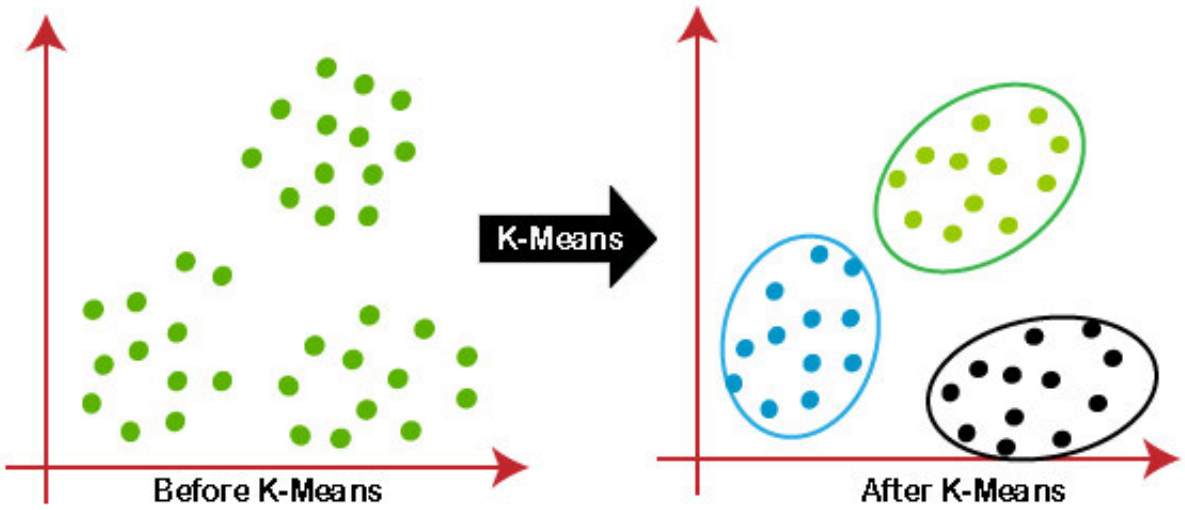


FIGURE 6. K-Means concept.

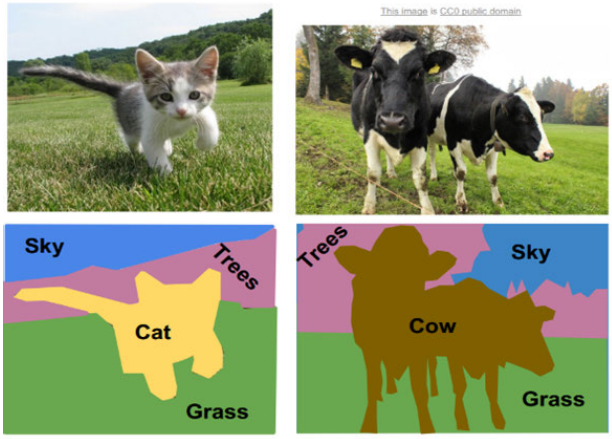


FIGURE 7. Image segmentation.

image on the left. This is why all of the seats are the same color. On the next tab, we've put together some examples that talk about specific items, which is why each seat has a different colour.

This is how different strategies for dividing images work at different levels of complexity and give different levels of results. Image division is a vast topic, and there are many different ways to look at the cycle. We can put image division into the groups shown by the following lines:

1) **Approach-Based Classification:** In its most essential sense, image division is object ID. An algorithm can't arrange the various parts without first recognizing an item. From easy to confounded executions, all image division works based on object ID. Thus, we can order image division techniques based on transit algorithms to recognize objects, which means gathering comparative pixels and isolating them from unique pixels.
2) **Technique-Based Classification:** We utilize these techniques as indicated by the kind of image we need to process and dissect and the kind of results we need to get from it. These algorithms expect you to have the underlying information of the image you are utilizing. This incorporates the pixels, appropriations, histograms, pixel thickness, shading circulation, and other important data. Then, at that point, you should have the primary information on the area you need to isolate from the image."

VI. PROPOSED ALGORITHM

To develop an algorithm for plant severity examination using disease detection on leaves, incorporating clustering and neural networks following the general steps outlined below:

1) **Data collection and preprocessing:** Gather a dataset of labeled images of healthy and diseased plant leaves. Preprocess the images by resizing, normalizing, and augmenting them if necessary.
2) **Feature extraction:** Extract relevant features from the preprocessed images. This step can involve various techniques, such as extracting texture features and color histograms or using pre-trained convolutional neural networks (CNNs) for feature extraction.
3) **Clustering:** Apply a clustering algorithm, such as k-means or DBSCAN, to group similar features together. This step helps identify patterns and separate healthy and diseased leaves based on their extracted features.
4) **Neural network training:** Use the clustered dataset to train a neural network model. This model can be a CNN, which is well-suited for image classification tasks. You can either train the CNN from scratch or use transfer learning by fine-tuning a pre-trained

CNN, such as VGG16, Inception, or ResNet, with your dataset.

5) **Model evaluation:** Evaluate the trained model using appropriate metrics, such as accuracy, precision, recall, and F1-score. Split your dataset into training and testing sets to assess the model's performance on unseen data. Adjust hyperparameters and model architecture if necessary to improve performance.
6) **Severity assessment:** Once the model is trained and evaluated, you can use it to predict the severity of diseases on new, unseen plant leaves. The model will classify the leaves into healthy or diseased categories, and you can determine the severity based on the predicted class probabilities or confidence scores.

It's worth mentioning that the success of the algorithm depends on the quality and diversity of the dataset, the choice of the clustering algorithm, and the effectiveness of the neural network model. Experimentation and iteration may be necessary to optimize the algorithm and achieve accurate disease detection and severity assessment. The proposed algorithm operates as follows:

1) select_leaf_image():
   a) This function prompts the user to select a leaf image for examination.
   b) Returns the selected leaf image.
2) enhance_image(image):
   a) This function enhances the contrast of the input image and resizes it for better clarity.
   b) Returns the enhanced image.
3) perform_color_segmentation(image):
   a) This function performs colour image segmentation on the input image.
   b) Returns the segmented image.
4) apply_kmeans_segmentation(image):
   a) This function applies K-means clustering to the input image for segmentation.
   b) Returns the segmented image.
5) convert_to_lab(image):
   a) This function converts the input image from RGB color space to Lab* color space.
   b) Returns the image in Lab colour space.
6) create_training_set():
   a) This function creates a training set for image identification using a Multiclass SVM.
   b) Loads the training data set (Training_Data.mat).
   c) Returns the training set.
7) classify_disease(image, training_set):
   a) This function classifies the disease in the given image using the trained SVM model.
   b) Returns the identified disease category.
8) *main*():
   a) Main function to execute the algorithm steps in sequence.
   b) Calls the above functions in the following order:
   c) leaf_image = select_leaf_image()
   d) enhanced_image = enhance_image(leaf_image)
   e) segmented_image = perform_color_ segmentation(enhanced_image)
   f) kmeans_segmented_image = apply_kmeans_ segmentation(segmented_image)
   g) lab_image = convert_to_lab (kmeans_segmented_ image)
   h) training_set = create_training_set()
   i) disease_category = classify_ disease(lab_image, training_set)
   j) Print the identified disease category.
9) *main*():
   a) Call the main function to start the execution.

A. PROPOSED ALGORITHM CONCEPT

The proposed algorithm offers a systematic approach to automated plant disease detection by integrating various image processing techniques and machine learning methods. It begins by prompting the user to select a leaf image for examination, followed by enhancing the image's contrast and resizing it for clarity. Color segmentation is then performed to isolate regions of interest, and K-means clustering is applied to refine the segmentation further. The image is then converted to Lab color space for a better representation of color information. Subsequently, a training set is created using a Multiclass Support Vector Machine (SVM) to facilitate disease identification. The algorithm then utilizes the trained SVM model to classify the disease in the given image. Finally, the identified disease category is printed, providing actionable information for disease management. By automating the process of disease detection through image analysis and machine learning, this algorithm offers a valuable tool for farmers and agricultural professionals, enabling early detection and intervention to mitigate the impact of plant diseases on crop yield and food security, as shown in Table 10.

VII. IMPLEMENTATION AND RESULT ANALYSIS

This project employed MATLAB-based image processing tools to implement an algorithm for examining plant severity via disease detection on leaves using clustering and neural networks. The software environment encompasses MATLAB, augmented by the Image Processing Toolbox for essential image manipulation and analysis tasks. Efficient processing of image data, particularly for tasks involving large datasets or complex algorithms, necessitates a computer with a minimum of eight GB RAM and an Intel i5 processor. High-resolution displays and color-calibrated monitors facilitate accurate visualization and interpretation of image results. Ensuring that both hardware and software configurations are optimized for image processing tasks is critical to achieving efficiency and accuracy in the outcomes.

TABLE 1. Hyperparameters of the proposed approach.

Function	Hyperparameter	Description
perform_color_segmentation	number of segments	Specifies the desired number of segments for color image segmentation using algorithms like K-means.
apply_kmeans_segmentation	number of clusters	Determines the number of clusters for K-means clustering, influencing the segmentation granularity.
create_training_set	Kernel type (SVM)	Select the type of kernel function for the Support Vector Machine (SVM), affecting classification.
classify_disease	C (SVM regularization)	Controls the trade-off between achieving a low training error and minimizing the complexity of the model.
classify_disease	Gamma (SVM kernel coef.)	Influences the extent of the influence of a single training example, affecting decision boundary smoothness.

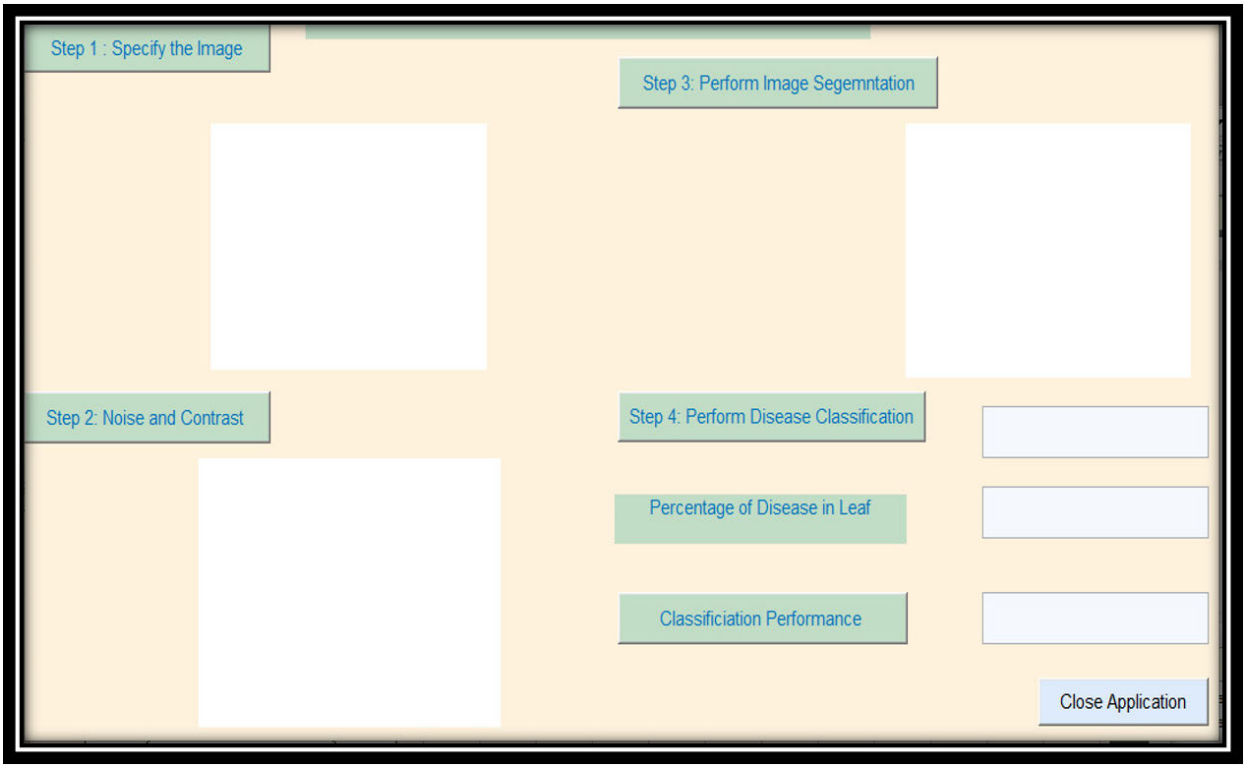


FIGURE 8. Implementation.

In the fig 8 shows the implementation of the proposed work. To run the implementation, the Detect Disease implementation file is opened in Matlab2015 a and clicks on the run button, the interface which is shown in fig 8. The implementation is based on classifying the diseases that are on the leaf or finding a healthy leaf. These diseases include Alternaria-Alternata, Anthracnose, Bacterial-Based Blight, Cercospora-Based Leaf Spot, and the Healthy Leaf.

Fig 9 shows that the image which is selected as the query image and for the analysis purpose, the image contrast is enhanced.

Fig 10 shows the segmentation of the images in which the segmentation and the selection of the images are done on the basis of K-Means clustering and, in this step, will select the cluster that is best suited for the examination purpose. As the clustering process is done, now the features extraction process continues, and fig 10 extracts the features related to entropy, variance, energy, etc.

Fig 11 shows the classification of the leaf image according to the disease identified in the image. This will display the name based on the dataset-based analysis done using the Multiclass SVM and the affected region of the image.

Fig 12 shows the accuracy of the classification of the image; for this, the 500 iterations are being carried out to find the percentage accuracy of the classification performed.

VIII. RESULT ANALYSIS

A. ALTERNARIA ALTERNATA DETECTION

Detecting Alternaria alternata, a common plant pathogenic fungus shown in fig 14, can be achieved using various approaches. Here is a proposed method using image analysis and machine learning.

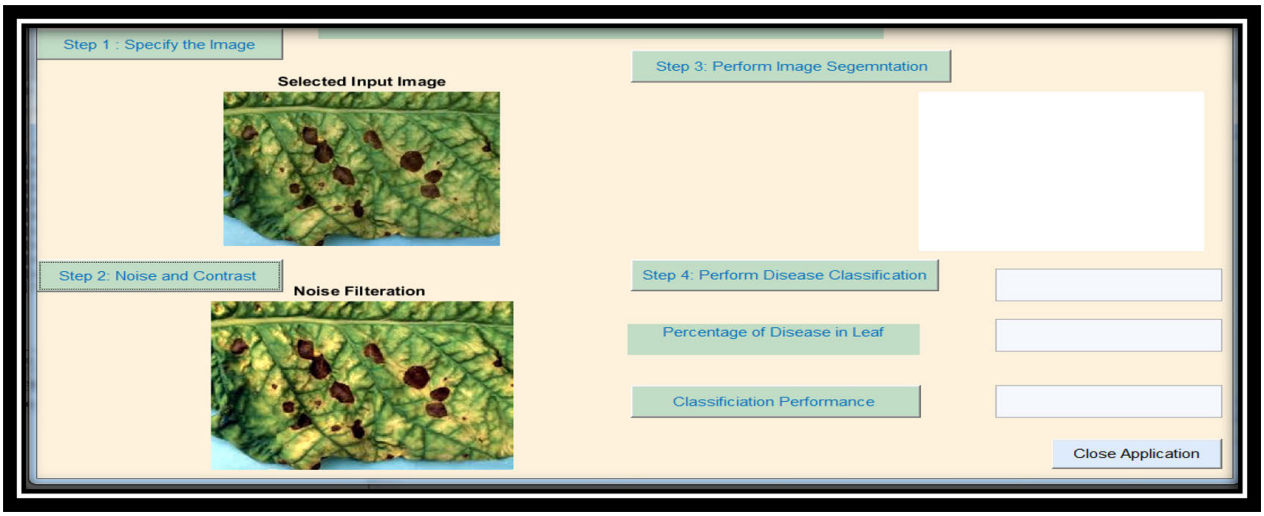


FIGURE 9. Query image selection and contrast enhancement.

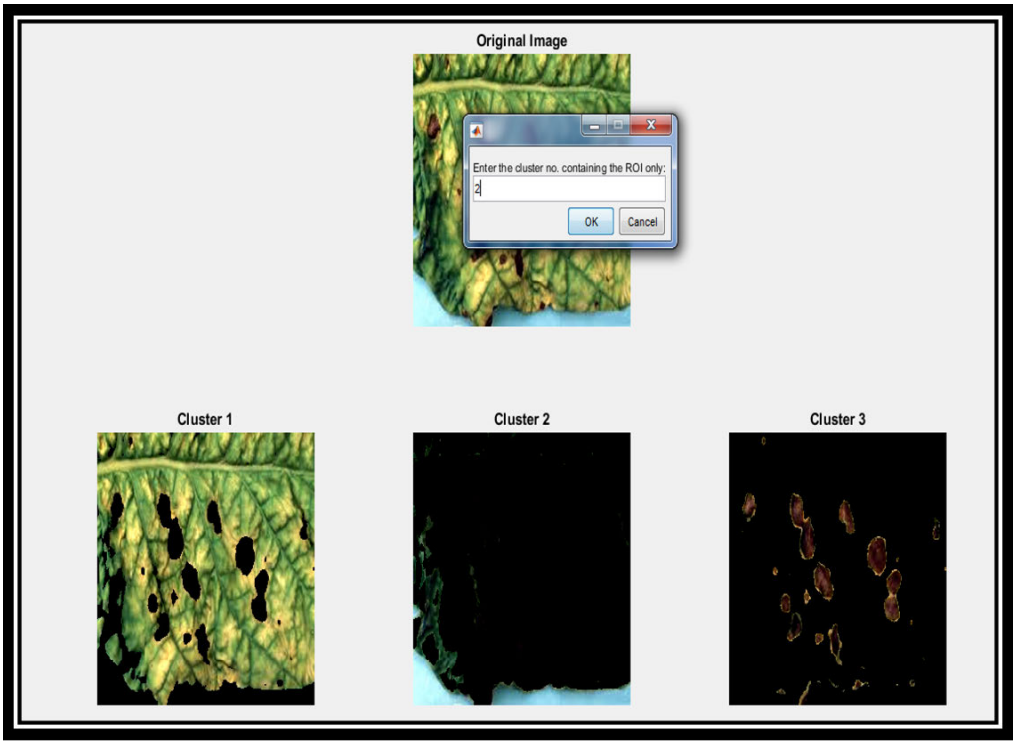


FIGURE 10. Segmentation.

The base paper which is taken is Bhagat et al. 2020 [34], Plant Leaf Disease Classification Using Grid Search Based SVM the accuracy which is achieved in plant disease identification is 84%.

According to the information in table 3 that was provided, the classification algorithm managed to attain a recommended accuracy of 84% when it came to the detection of Alternaria alternata. In actual practice, a level of accuracy of 96.7742% was reached.

The fact that the algorithm managed to reach an accuracy of 96.7742% demonstrates that it did an excellent job of reliably detecting Alternaria alternata in the dataset that was provided or in real-world circumstances. It achieved a higher level of accuracy than the estimated 84%, demonstrating that the algorithm is successful in determining instances of an infection caused by Alternaria alternata.

When evaluating the overall performance of the model, it is recommended that in addition to accuracy, other metrics such

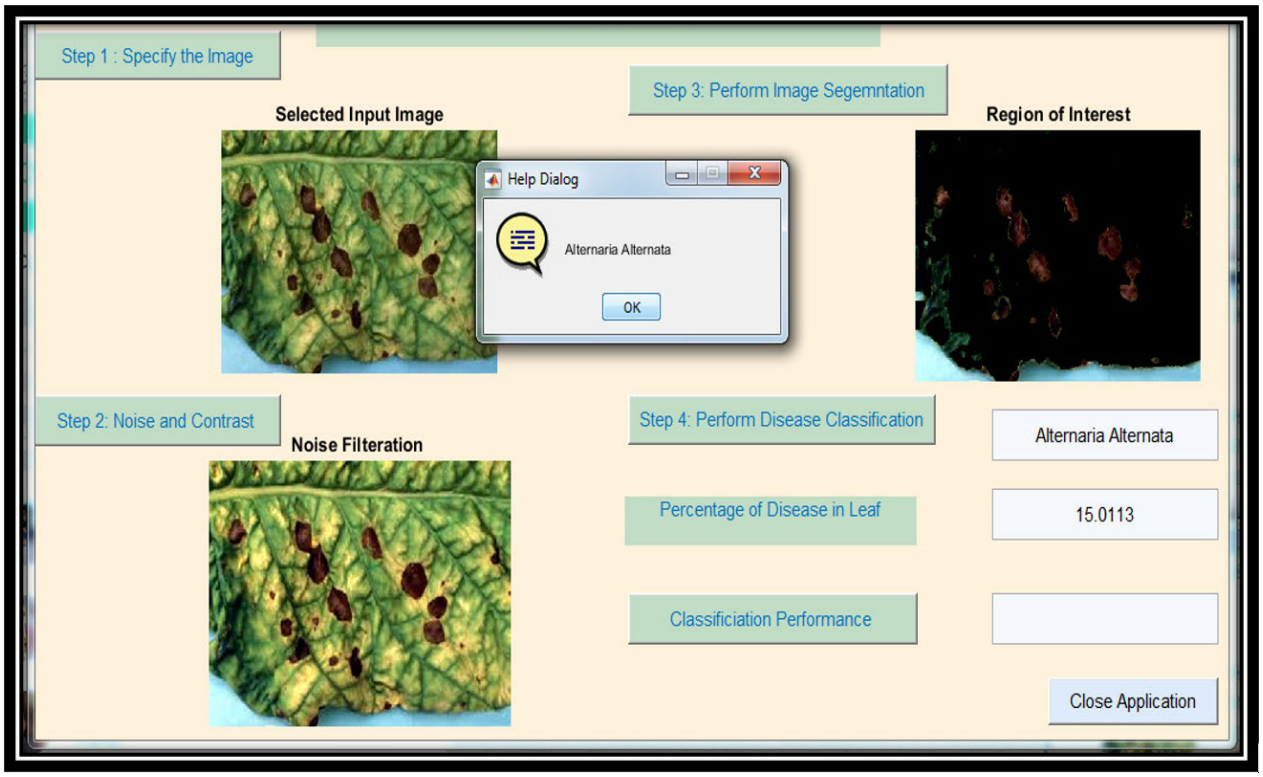


FIGURE 11. Classification.

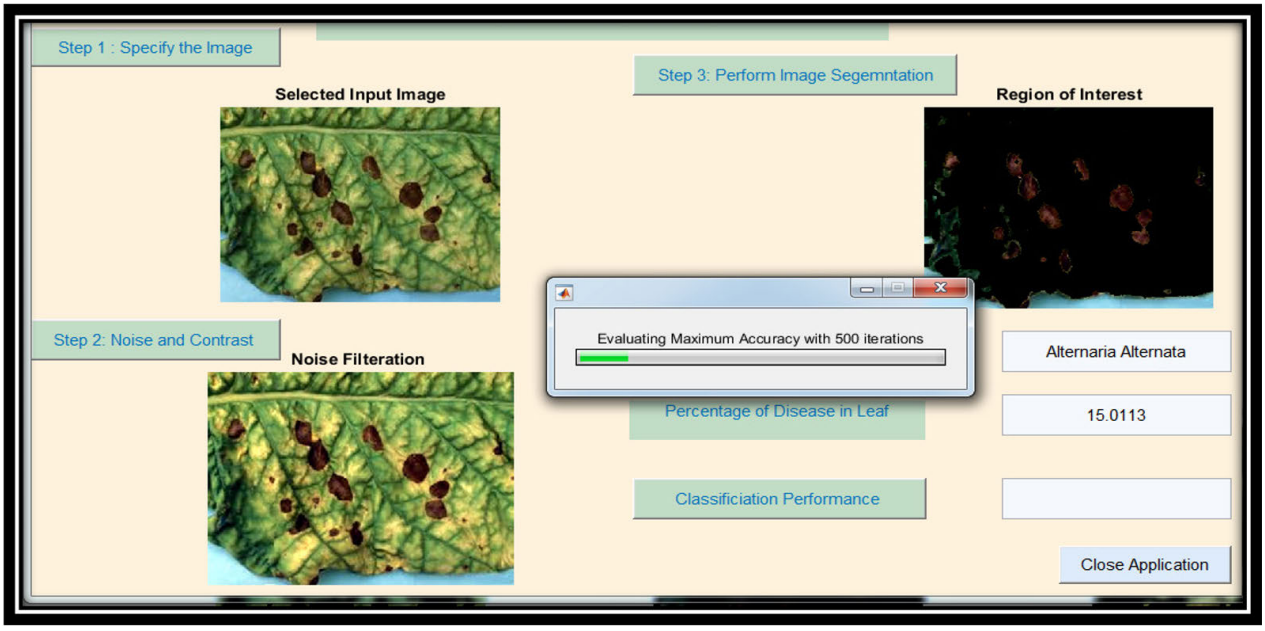


FIGURE 12. Percentage accuracy analysis.

as precision, recall, and F1-score be taken into consideration. It is essential to keep in mind that accuracy is just one of the evaluation measures.

It is advised that the algorithm be tested on a variety of datasets that contain a variety of plant species, disease severities, and environmental circumstances. This will further validate the accuracy and dependability of the method. This will assist in evaluating its powers of generalisation and ensure that it is effective across a variety of settings.

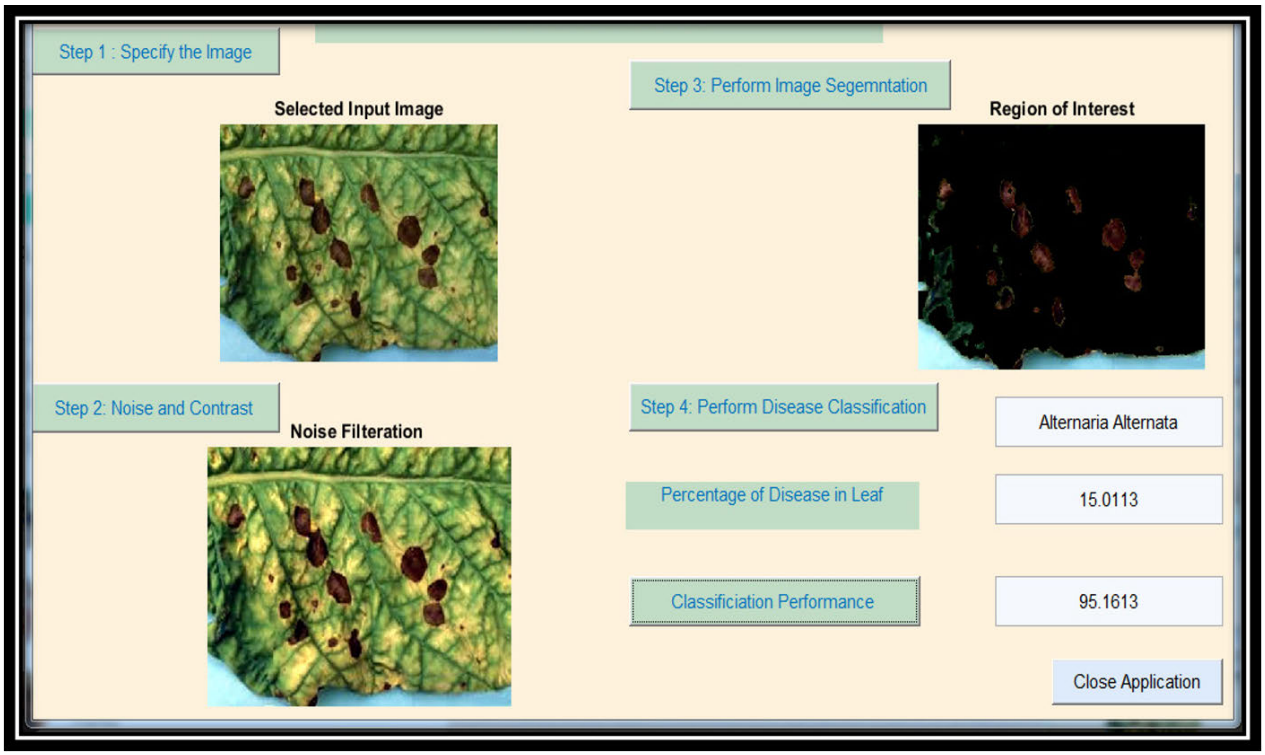


FIGURE 13. Percentage accuracy listing.



FIGURE 14. Test sample 1 alternaria alternata.

TABLE 2. Accuracy table for alternaria alternata.

Sample Number	Affected Region Identified %	Accuracy Achieved %
1	15.019	95.1613
2	16.214	98.3871
3	15.314	96.7742
4	15.85	96.7742
5	14.1935	96.7742
Average Accuracy		96.7742

B. ANTHRACNOSE

Anthracnose is a plant disease caused by various species of fungi, including Colletotrichum spp. and Glomerella spp. It affects a wide range of plants, including fruits, vegetables, trees, and ornamental plants, as shown in the fig. 16. To detect and manage anthracnose, you can utilize the following approaches:

According to the provided information in table 5, the classification algorithm achieved a proposed accuracy of

TABLE 3. Accuracy table for alternariaalternata comparison with base.

Classification Result	Base Percentage %	Proposed Accuracy Achieved %
Alternaria alternata	84%	96.7742%

TABLE 4. Accuracy table for anthracnose.

Sample Number	Affected Region Identified %	Accuracy Achieved %
1	17.8648	96.7742
2	15.7728	96.7742
3	15.0217	96.7742
4	23.6869	98.3871
5	15.0016	98.3871
Average Accuracy		97.41936

TABLE 5. Accuracy table for anthracnose comparison with base.

Classification Result	Base Percentage %	Proposed Accuracy Achieved %
Alternaria alternata	84%	97.41936%

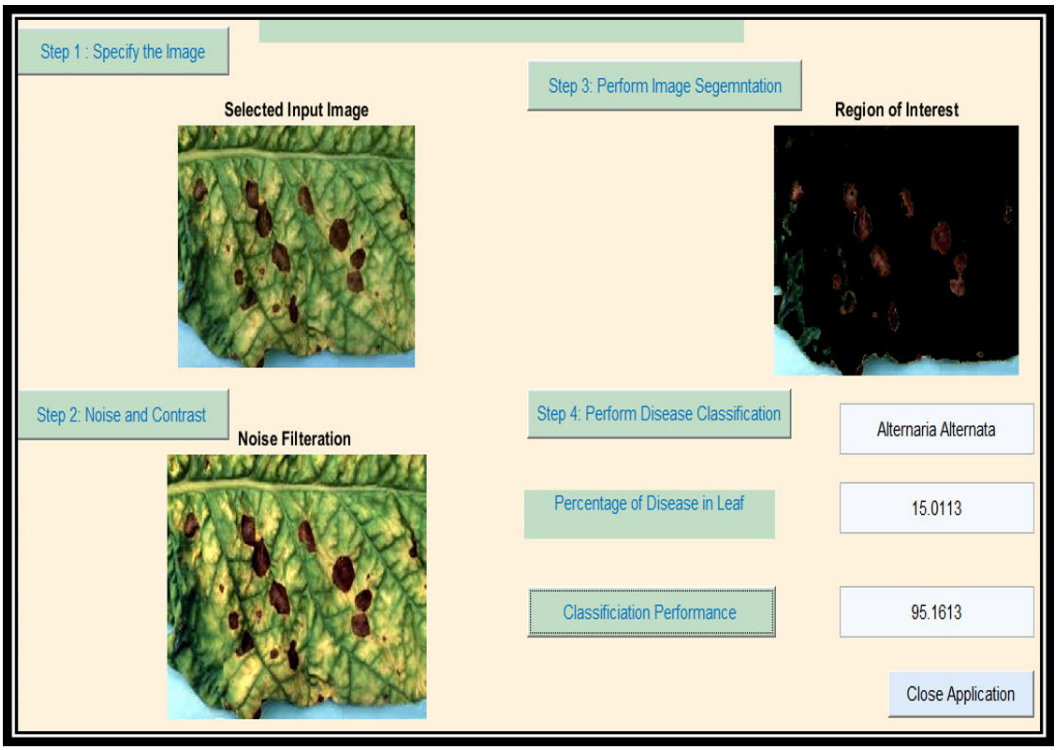


FIGURE 15. Classification result and accuracy result for sample 1 alternaria alternata.



FIGURE 16. Leaf sample 1 for the anthracnose.

84% for Alternaria alternata detection. However, the achieved accuracy in practice was 97.41936%.

This suggests that the algorithm performed substantially better than was initially anticipated, achieving a higher level of accuracy than was proposed. The fact that the algorithm managed to obtain an accuracy of 97.41936 per cent shows that it is helpful in accurately detecting Alternaria alternata within the provided dataset or in real-world circumstances.

In the same vein as the previous response, it is essential to take into account additional evaluation metrics such as precision, recall, and F1-score to acquire a more in-depth comprehension of the performance of the model.

Also, testing the algorithm on different datasets and figuring out how well it can be generalized will give us more information about how accurate and reliable it is in various situations and with varying species of plant that Alternaria alternata affects.

TABLE 6. Accuracy table for bacterial blight.

Sample Number	Affected Region Identified %	Accuracy Achieved %
1	15.0077	96.7742
2	15.0062	96.7742
3	15.0024	98.2
4	15.0142	96.7742
5	15.0016	98.3871
Average Accuracy		97.41936

C. BACTERIAL BLIGHT

According to table 7, the classification algorithm achieved a proposed accuracy of 84% for anthracnose detection. However, the achieved accuracy in practice was 97.13065%.

This suggests that the algorithm performed noticeably better than was previously predicted, surpassing the accuracy that was proposed. The fact that the algorithm was able to obtain an accuracy of 97.13065 per cent shows that it is capable of accurately detecting anthracnose in the dataset that was provided or in real-world circumstances.



FIGURE 17. Classification result and accuracy result for sample 1 anthracnose.



FIGURE 18. Test sample 1 for bacterial blight.

It is essential to keep in mind that accuracy may provide a partial picture of the performance of the model. It is recommended that other evaluation measures such as precision, recall, and F1-score be taken into consideration when determining how effective the model is in detecting anthracnose as a whole.

In addition, it is vital to validate the performance of the algorithm on a variety of datasets and to take into consideration its potential to generalise. A more in-depth comprehension of the algorithm's precision and dependability can be attained by the performance of additional testing and analysis in a broader variety of use cases and data sets.

TABLE 7. Accuracy table for bacterial blight comparison with base.

Classification Result	Base Percentage %	Proposed Accuracy Achieved %
Anthracnose	84%	97.13065%

IX. DISCUSSION

In the proposed research, we introduced an integrated approach that combines advanced image processing techniques and machine learning algorithms to address the critical issue of automated plant disease detection and classification. Our methodology involved several key steps, including image enhancement, color segmentation, clustering, and classification using a Multiclass Support Vector Machine (SVM) with grid search optimization. Through this approach, we aimed to provide a systematic framework for identifying diseased regions within plant leaf images and accurately classifying the associated diseases.

One of the primary findings of our study was the importance of preprocessing techniques, such as image enhancement and color segmentation, in improving the quality of input data for disease classification. By enhancing image contrast and isolating regions of interest, we were able to reduce noise and enhance the clarity of diseased areas, leading to more accurate disease classification results. Additionally, the application of clustering algorithms, particularly K-means clustering, facilitated the identification of distinct clusters within the segmented images, further enhancing the precision of disease localization.

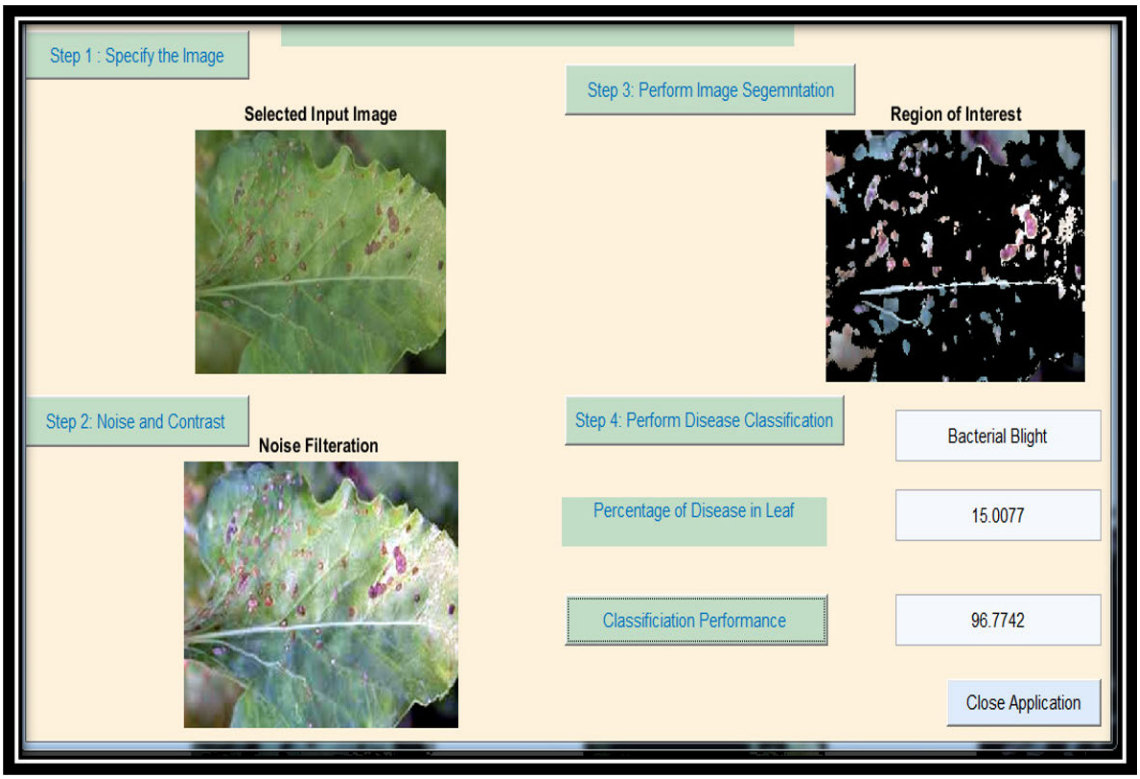


FIGURE 19. Classification result and accuracy result for sample 1 bacterial blight.

TABLE 8. Approaches comparison [35], [36].

Aspect	Proposed Approach	Base Approach
Methodology	Integrates image processing and SVM classification	Relies on SVM classification with Grid Search optimization
Disease Detection	Detects specific diseases through image analysis	Classifies leaves as healthy or unhealthy
Computational Efficiency	Utilizes clustering and segmentation for efficiency	Focuses on optimizing SVM parameters for accuracy
Complexity	Involves multiple steps including segmentation and SVM	Emphasizes SVM parameter tuning for classification
Flexibility	Offers flexibility through image processing techniques	Relies on SVM's inherent flexibility in classification
Accuracy	Accuracy influenced by image processing and SVM tuning	Accuracy dependent on SVM parameter optimization
Eco-Friendliness	Potential reduction in pesticide use through early detection	No direct eco-friendliness component mentioned

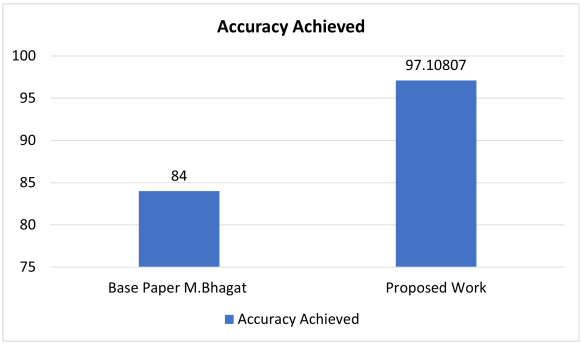


FIGURE 20. Accuracy achieved.

Furthermore, our research demonstrated the effectiveness of the Multiclass SVM classifier in accurately distinguishing between different types of plant diseases based on the extracted features from the segmented images. By optimizing SVM hyperparameters using the grid search technique, we were able to identify the optimal configuration that maximized classification accuracy. This highlights the importance of fine-tuning machine learning models to achieve optimal performance in disease classification tasks. However, it is essential to acknowledge some limitations of our approach. While our methodology showed promising results in controlled experimental settings, further validation on diverse datasets and real-world scenarios is necessary to assess its robustness and generalizability. Additionally, the computational complexity of specific image processing techniques and the training process of SVM models may pose scalability challenges, particularly for large-scale applications.

In conclusion, our study contributes to the growing body of research aimed at leveraging technology to address challenges in agriculture, particularly in the context of plant disease management. By integrating image processing and machine learning techniques, we have developed a viable

solution for automated disease detection and classification, with the potential to enhance agricultural sustainability and productivity. Moving forward, continued research and collaboration are essential to refining and deploying these methods of practical farm settings, ultimately benefiting farmers and contributing to global food security.

Limited Dataset Size: The study may be constrained by the size and diversity of the dataset used for training and testing the proposed approach. A small or biased dataset may limit the generalizability of the results and could lead to overfitting or underperformance on real-world data.

X. LIMITATIONS OF STUDY

1) Dependency on Image Quality: The effectiveness of the proposed approach heavily relies on the quality of input leaf images. Variations in image resolution, lighting conditions, and image artefacts could impact the accuracy of disease detection and classification.
2) Sensitivity to Parameter Tuning: The performance of machine learning algorithms, such as Support Vector Machines (SVM), is sensitive to hyperparameter tuning. Suboptimal selection of parameters or biases in the grid search optimization process could affect the robustness and reliability of the classification model.
3) Computational Resource Requirements: Implementing the proposed approach may require significant computational resources, including processing power and memory. This could pose challenges for researchers or practitioners with limited access to high-performance computing infrastructure.
4) Scope Limitation: The proposed approach focuses primarily on the automated detection and classification of plant diseases based on leaf images. However, it may overlook other important factors contributing to disease development and spread, such as environmental conditions, genetic factors, and agronomic practices.
5) Evaluation Metrics: The study may lack comprehensive evaluation metrics to assess the performance of the proposed approach accurately. Limited metrics or incomplete validation procedures could obscure the true efficacy of the method and its comparative advantages over existing techniques.

Addressing these limitations and conducting further research to refine the proposed approach will be essential for enhancing its effectiveness and applicability in real-world agricultural settings.

XI. CONCLUSION

The K-Means algorithm is used to look at infected leaves and figure out where the infection is most common. The system does an excellent job of telling the difference between infected and healthy leaves. A multiclass SVM-based classifier is used to put diseases into groups. This classifier uses a database of information made from the analysis of many leaves. Diseases like Alternaria Alternata, Anthracnose, Bacterial Blight, and Cercospora Leaf Spot can be correctly identified by the system. The system works better than older methods because it is about 96–97% accurate.

XII. FUTURE SCOPE

The future scope of plant severity examination using disease detection on leaves using clustering and neural networks is promising and encompasses several potential advancements. Here are a few areas that hold significant potential for future development:

1) **Improving accuracy and robustness:** Future research can focus on enhancing the accuracy and robustness of disease detection algorithms. This can involve exploring more advanced clustering algorithms that can better identify patterns and separate healthy and diseased leaves. Additionally, advancements in neural network architectures and techniques such as attention mechanisms, ensemble learning, or adversarial training can contribute to improved disease detection performance.
2) **Multiclass and multi-disease detection:** Extending the algorithm to handle multiple diseases simultaneously and classify leaves into multiple disease categories would be valuable. This can involve training the neural network to recognize and differentiate between various diseases, allowing for more comprehensive plant health assessments.
3) **Real-time monitoring and mobile applications:** Developing real-time disease detection systems that can operate on mobile devices or embedded platforms would enable on-site monitoring and immediate decision-making. This can involve optimizing the algorithm for efficiency and adapting it to run in resource-constrained environments.
4) **Transfer learning and domain adaptation:** Investigating the use of transfer learning techniques can be beneficial for leveraging pre-trained models and knowledge from related domains. Adapting models trained on one plant species to detect diseases in other species can reduce the need for large labeled datasets and improve generalization across plant types.
5) **Integration with precision agriculture technologies:** Integrating disease detection algorithms with other precision agriculture technologies, such as remote sensing, drones, or IoT sensors, can enable comprehensive monitoring of plant health at larger scales. This integration can provide valuable insights for targeted interventions and resource management in agriculture.
6) **Explainability and interpretability:** Enhancing the interpretability of disease detection algorithms can help build trust and understanding among users. Researching methods to explain the model's decisions and provide insights into the features influencing disease detection can facilitate adoption in practical applications.
7) **Long-term monitoring and disease forecasting:** Expanding the scope to include long-term monitoring

and disease forecasting can enable proactive disease management. By integrating historical data, environmental factors, and disease progression patterns, predictive models can provide early warnings and recommend preventive measures to minimize crop damage.

8) **Data sharing and collaborative platforms:** Establishing data-sharing platforms or repositories can encourage collaboration and facilitate the development of more robust disease detection algorithms. Sharing annotated datasets and benchmarking results can foster innovation and drive advancements in the field.

These future directions have the potential to enhance the accuracy, efficiency, and practicality of plant severity examination using disease detection on leaves. Continued research and collaboration among plant pathologists, data scientists, and agricultural stakeholders can drive the development and adoption of these advancements.

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